Thomas C. Martinez

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Research Interests

Algebraic combinatorics; total positivity and positroid varieties; Coxeter-Catalan combinatorics

Education

University of California, Los Angeles **2022–present**
PhD, Mathematics (expected 2027) · Advisor: Pavel Galashin Los Angeles, CA

Harvey Mudd College **2017–2021**
BS, Mathematics · High Distinction and Honors in Mathematics Claremont, CA
Thesis: *The Slice Rank Polynomial Method* · Advisor: Mohamed Omar

Publications and Preprints

Preprints

[P1] (with M. J. Tyler) *Toric Richardson Varieties and Slice Links.* 2026. arXiv:2609.28536.

[P2] *The Combinatorics of Affine Deodhar Diagrams.* 2026. arXiv:2607.15672.

[P3] (with B. B. Connelly*, E. Ito*, O. Shevchenko, and K. Yang*) *Unimodality of q-Fibonomial Coefficients for Small Cases.* 2026. arXiv:2605.12822.

Manuscripts in Preparation

[W1] *Parking on Positroids.* Manuscript in preparation.

Publications

[1] (with T. Jaklitsch, S. J. Miller, and S. Mukherjee) *Connections of Class Numbers to the Group Structure of Generalized Pythagorean Triples. Rocky Mountain Journal of Mathematics* **55**(2), 369–380, 2025.

[2] (with T. Blum, C. Choi, A. Hoey, J. Iskander, and K. Lakein) *On Class Numbers, Torsion Subgroups, and Quadratic Twists of Elliptic Curves. Transactions of the American Mathematical Society* **375**(1), 351–368, 2022.

[3] (with H. V. Chu, D. King, N. Luntzlara, S. J. Miller, L. Shao, C. Sun, and V. Xu) *Generalizing the Distribution of Missing Sums in Sumsets. Journal of Number Theory* **239**, 402–444, 2022.

[4] (with S. J. Miller, C. Mizgerd, J. Murphy, and C. Sun) *Generalizing Zeckendorf's Theorem to Homogeneous Linear Recurrences, II. Fibonacci Quarterly* **60**(5), 220–230, 2022.

[5] (with S. J. Miller, C. Mizgerd, and C. Sun) *Generalizing Zeckendorf's Theorem to Homogeneous Linear Recurrences, I. Fibonacci Quarterly* **60**(5), 231–254, 2022.

[6] (with F. Beretta, J. Dimino, W. Fang, S. J. Miller, and D. Stoll) *On Benford's Law and the Coefficients of the Riemann Mapping Function for the Exterior of the Mandelbrot Set. Fractal and Fractional* **6**(10), 534–552, 2022.

[7] (with A. T. Benjamin and J. Lentfer) *Counting on Euler and Bernoulli Number Identities. Fibonacci Quarterly* **58**(5), 30–33, 2020.

* Undergraduate collaborator

Selected Talks and Presentations

UC San Diego Combinatorics Seminar **November 2026**
Affine Patches of Open Positroid Varieties San Diego, CA

UC Davis Algebra and Discrete Mathematics Seminar **November 2026**
Affine Patches of Open Positroid Varieties Davis, CA

UT Dallas Algebra and Combinatorics Seminar **November 2026**
Affine Patches of Open Positroid Varieties Dallas, TX

SIAM TXLA 2026: Algebraic and Geometric Structures in Discrete Mathematics Session **October 2026**
Two Ways to Park on Positroids Dallas, TX

CMND 2026: Algebraic Combinatorics and Applications **July 2026**
Lightning talk: *Affine Patches of Open Positroid Varieties* Notre Dame, IN
Binghamton University Combinatorics Seminar **March 2026**
The Combinatorics of Affine Deodhar Diagrams Binghamton, NY
Cornell Discrete Geometry and Combinatorics Seminar **February 2026**
The Combinatorics of Affine Deodhar Diagrams Ithaca, NY
CanaDAM 2025: Algebraic Combinatorics Minisymposium **May 2025**
Invited minisymposium talk: *Affine Deodhar Diagrams and Rational Dyck Paths* Ottawa, ON
LatMath 2025 **March 2025**
Poster: *Affine Deodhar Diagrams and Rational Dyck Paths* (Outstanding Poster Award) Los Angeles, CA
UCLA Combinatorics Learning Seminar **June 2024**
The (Strong and Weak) Bruhat Order Los Angeles, CA
UCLA Combinatorics Research Seminar **April 2024**
Complex Weighted Tilings and Kasteleyn's Theorem Los Angeles, CA
UCLA Combinatorics Learning Seminar **April 2024**
Introduction to Root Systems and Reflection Groups Los Angeles, CA
UCLA Combinatorics Research Seminar **February 2024**
Deograms and Dyck Paths Los Angeles, CA
UCLA Combinatorics Research Seminar **May 2023**
The Slice Rank Polynomial Method Los Angeles, CA
19th International Fibonacci Conference **July 2020**
Generalizing Zeckendorf's Theorem to Homogeneous Linear Recurrences Virtual
University of Virginia REU Symposium **July 2020**
On Class Numbers, Torsion Subgroups, and Quadratic Twists of Elliptic Curves Virtual
Young Mathematicians Conference **August 2019**
Poster: *Generalizing the Distribution of Missing Sums in Sumsets* Columbus, OH
UMass Amherst REU Conference **July 2019**
Generalizing the Distribution of Missing Sums in Sumsets Amherst, MA

Honors/Awards

LatMath Outstanding Poster Award **2025**

Raymond Redheffer Prize **2025**
UCLA departmental award for excellence in calculus teaching.

Greever Research Prize – Honorable Mention **2021**
Harvey Mudd departmental award for original mathematical research.

Alvin White Prize **2020**
For contributions to the Harvey Mudd mathematics community.

Teaching

Teaching Assistantships **University of California, Los Angeles**
- Math 32A: Multivariable Differential Calculus [x3] 2023–2025
- Math 61: Introduction to Discrete Structures [x7] 2022–2025
- Math 184: Enumerative Combinatorics Spring 2025
- Math 180: Graph Theory Fall 2024
- Math 115A: Linear Algebra Winter 2024
- Math 31B: Integration and Infinite Series Winter 2024

Teaching Assistantships **Cornell University**
- Math 2930: Differential Equations for Engineers Spring 2026

Tutoring **Harvey Mudd College**
- *Academic Excellence* Peer Tutor Fall 2019 - Spring 2021
  Proof-based Calculus, Linear Algebra, Differential Equations, Multivariable Calculus, Discrete Mathematics

Conference, Seminar, and Committee Organization

- UCLA Directed Reading Program Committee **September 2025–present**
- 21st International Fibonacci Conference (Harvey Mudd College – Claremont, CA) **July 2024**
- UCLA Combinatorics Learning Seminar (Coxeter Groups) **Spring 2024**
- UCLA Combinatorics Research Seminar **Spring 2024**
- UCLA Combinatorics Research Seminar **Winter 2024**
- 20th International Fibonacci Conference (University of Sarajevo – Sarajevo, Bosnia & Herzegovina) **July 2022**
- 19th International Fibonacci Conference (Virtual, via Zoom) **July 2020**

Leadership / Outreach

UCLA Directed Reading Program **University of California, Los Angeles**
Mentor for semester-long undergraduate reading projects September 2023–present

- Yuyang Zhang: Polyhedral Combinatorics Fall 2026
- Brendan Connelly, Ezekiel Ito, Kacey Yang (with Olha Shevchenko): q-Fibonomial coefficients Spring 2026
- Arjun Mittha: Combinatorial optimization Spring 2025
- Darsh Verma: The Art of Combinatorial Proof Spring 2025
- Thomas Slavonia: Schubert polynomials and pipe dreams Spring 2024
- Thomas Slavonia: Representation theory of the symmetric group Winter 2024
- Anika Misra: Graph theory Winter 2024
- Akash Kote: The probabilistic method in combinatorics Fall 2023

UCLA Learning Assistant Program **University of California, Los Angeles**
Math Learning Assistant Coordinator September 2024–September 2025

- Facilitated discussions on mathematical pedagogy and resolved course-specific content questions.
- Oversaw the Learning Assistant Program, promoting collaborative learning in weekly discussion sections.

The Motivational Edge **Allapattah, FL**
Volunteer Tutor September 2021–March 2022

- Provided weekly tutoring in mathematics, English, and science to elementary and middle school students.
- Collaborated with directors to develop personalized learning plans based on student progress and needs.

City Year Miami **Miami Jackson Senior High School**
Team Leader and Math AmeriCorps Member August 2021–June 2022

- Led and managed a team of four AmeriCorps members, delivering academic tutoring, behavioral interventions, and school-wide initiatives to over 200 ninth- and tenth-grade students.
- Provided targeted support in Algebra 1 to 71 students, focusing on attendance, behavior, and academic improvement.
- Designed and implemented weekly interventions for 11 students to address critical academic and behavioral challenges.

Number Theory & Benford's Law Research Groups **Polymath Jr. Program (Virtual)**
Assistant Director June 2021–August 2021

- Co-mentored 40 students guiding them through advanced research topics.
- Organized and facilitated a weekly seminar series for students to present research progress and receive feedback.
- Contributed to the production of several publications and conference presentations by program participants.

Languages

Written/Spoken:

- English (native) · Portuguese (native) · Spanish (fluent) · French (basic)

Last updated: October 20, 2026